

SHOWCASE

Sign language translator

A group of researchers from the University of California, LA, have designed a device that packs some cool tech within a glove form factor. This glove tracks finger movements in real-time, allowing it to translate sign language.

The glove-like wearable uses machine learning tech to bridge the distance between signers and non-signers with a sign-to-speech translation system. The signer wears the glove, and each finger on the glove has stretchable sensors made of electrically conducting yarns. These sensors pick up hand movements and finger positions, allowing for interpretation of letters, numbers, words and more.



Each movement is converted into electrical signals before being sent to a tiny circuit board on the

wrist. This board then relays those signals to a smart device wirelessly where they can be converted into words or text. Currently, the wearable can only translate at a speed of one word per second.

But that's not all, in order to fully understand sign language, which also includes facial expressions, researchers have included adhesive sensors, which are placed on the sides of the mouth and between the eyebrows. These are lightweight to the point where you won't even feel them.



Cradlewise smart crib

A common complaint among new parents is the lack of sleep. Understandable, a waking baby is usually a cranky baby. To that end, the Cradlewise cradle might be just what you need. This cradle can detect when a baby is waking and gently bounce it back to sleep.

The Cradlewise comes with a built-in night-vision camera and microphone. Using these it monitors the baby and over time, it's AI-based software learns a sleep profile for the baby, determining and scheduling bedtime and wake-up times.

If it detects that the baby is showing signs of waking up early, it gently starts bouncing up and down, in the same way that someone holding a baby might try to put them to sleep. The cradle also emits white noise from the integrated speakers. Once it has determined that the baby is asleep, the bouncing ceases and the noise is replaced with instrumental music.

Parents can choose the type of music they want played via an accompanying app. The app can also, of course, be used to monitor the baby and analyse sleep patterns and even just watch the baby, like a regular baby monitor.

Sony Reon Pocket

We're all too familiar with hot summers here, and can't wait to get back to our air-conditioned homes after some time spent outside. But what if you could carry your AC with you? Well, it's not like we're doing a lot of roaming around nowadays, but if we do, it would be nice to be cool on the go.

This is where Sony's Reon Pocket comes into play. Reon Pocket is a thermoelectric cooler that works via your smartphone with an accompanying app. Also, if you happen to be from cooler regions to begin with, it also doubles as a portable body heater.

The device can cool the wearer's body by 13 degrees, and raise body temperature by around 8 degrees when needed.

This Bluetooth device is about the size of a wallet, and fits into an undershirt pocket which is placed at the base of the neck. Once placed, with the help of an app, users



may control temperature via their smartphone. The device supports both iOS and Android devices, and has a battery life of about 90 minutes. The charge time is about 2 hours and it weighs roughly 85 grams. The Reon Pocket retails for around \$130, or around ₹9,700.